A separator for unequal witness intervals

Human Compression project: L044 continuation

All marks below are involutions satisfying string commutations. The restricted actions used in the first lemma need no intersection property.

Lemma 1 (Equality in the truncated count)**.** *Retain the actual local crossing setting of the truncated-orbit theorem, with active labels $m, \dots, y$, crossing colour j, and $j+1 \le x < y$. Let $L = Q_{<j}$, $R' = Q_{\{j+1,\dots,x-1\}}$ and $C_A = a^{LR'}$, $C_B = b^{LR'}$ for the crossing endpoints. Suppose*

$$h = j - m \ge 2, \quad v = x - j - 1 \ge 1, \quad |C_A| + |C_B| = h + v + 2.$$

If the first omission $Q_{>m}$ is intransitive, it has exactly two orbits: a singleton $\{z\}$ contained in $C_A \cup C_B$, and its transitive complement in the full local orbit W.

Proof. The left and truncated right active lengths have maxima h, v, exactly as proved in the complete truncated count. If $h + v \ge 4$, the sharper two-factor bound makes simultaneous attainment on the same side impossible: that side alone would require $h + v + 2$ points. The only remaining pair is $(h, v) = (2, 1)$. If both maxima occurred on one side, equality in the earlier bound would give $|C_A| = 4$, $|C_B| = 1$, after renaming. The single-involution factor R' cannot be transitive on four points. If both factors were intransitive, the orbit bound would give at least six points. Thus L is transitive on C_A. Every later right mark fixes a, b and centralizes L, so fixes C_A and C_B pointwise. Both sets would therefore be invariant under the full LR, making them the complete omission sides. No mark after $j+1$ could be active, contrary to $y > x = j + 2$. This excludes simultaneous attainment in the last case too. Rename the sides so that

$$|C_A| = h + 1, \qquad |C_B| = v + 1, \qquad L|_{C_A} = \mathrm{Sym}(C_A), \quad R'|_{C_B} = \mathrm{Sym}(C_B).$$

The left degree is at least three, so commutation makes R' fix C_A pointwise. Every further right mark fixes a, being at distance at least two from j. Thus the full right factor $R = Q_{>j}$ fixes a. Since LR is transitive on the actual side A, we have $A = a^L = C_A$, with R fixing all of A. If $v \ge 2$, the right degree is also at least three, and L fixes C_B pointwise.

When $v = 1$, the group R' is the transposition of $C_B = \{b, b_1\}$. Suppose a left mark exchanges these two points. It centralizes the full right factor. Every right mark after $j+1$ fixes b, so would also fix b_1. The first right mark preserves the pair, as does L by the definition of C_B. Thus the complete LR would preserve C_B, making $B = C_B$. The full right factor also fixes the other side A, as just proved. It would have no active label after $j+1$, contrary to $y > x = j + 2$. Hence L fixes C_B pointwise in this case as well. In either case L fixes b and centralizes R, so the other actual side is $B = b^R$, with L fixing all of B. The right action on B is transitive; it need not be symmetric.

Reverse the h left marks on A and use their exhausted anchored path $a_0 = a, \dots, a_h$. As in the complete minimum-orbit proof, the only possible extra transposition in those marks is $(a_{h-1}\ a_h)$ in the mark three positions from the end. If present, deleting the first mark ρ_m would leave the left orbit graph connected. The crossing and the transitive right action would then make $Q_{>m}$ transitive on W, a contradiction. Hence the remaining left marks give exactly $\mathrm{Sym}(A \setminus \{a_h\})$ and fix a_h.

For $h \ge 3$, ρ_j centralizes the pure $\mathrm{Sym}(\{a_1, \dots, a_h\})$, so fixes its support. For $h = 2$, its possible extra transposition $(a_1\ a_2)$ would again make the first omission transitive through its connected orbit graph; therefore it is absent. Thus $z = a_h$ is fixed by ρ_j and every mark of $Q_{>m}$.

The remaining left marks are transitive on $A \setminus \{z\}$, the right marks are transitive on B, and the crossing joins these sets. They give one transitive complementary orbit, as claimed. In particular $z \in A = C_A$. □

Lemma 2 (The short left sides)**.** *In the same actual local crossing setting, suppose $h = j - m$ is zero or one, and the first omission $Q_{>m}$ is intransitive. Let $R' = Q_{>j}$ and $v = y - j \geq 1$, or let $R' = Q_{\{j+1,\dots,x-1\}}$ and $1 \leq v = x - j - 1 < y - j$. Put $C_A = a^{LR'}$, $C_B = b^{LR'}$ as before. If $|C_A| + |C_B| = h + v + 2$, the first omission has one fixed point in $C_A \cup C_B$ and one transitive complementary orbit on W.*

Proof. The anchored action and strict prefix results give active right lengths v_A, v_B on these sets with maximum v. An orbit through its anchor under R' has at least $v_A + 1$ or $v_B + 1$ points, respectively.

If $h = 0$, the left factor is trivial. Equality forces one of these endpoint orbits to have $v + 1$ points and the other to be a singleton, with no active R'-mark there. Every remaining right mark, if there is one, fixes that other anchor because its index is at least $j + 2$. Thus one actual omission side is a singleton. The first omission is the full right factor and is transitive on the other side, proving the assertion.

Suppose $h = 1$, so L is generated by the single mark ρ_m. The full right factor $R = Q_{>j}$ cannot be transitive on both actual omission sides: otherwise $\langle \rho_j, R \rangle = Q_{>m}$ would be transitive on W. Name a side where R is intransitive as A. Since LR is transitive there and L has order two, R has precisely two orbits on A, exchanged by L. Consequently C_A is a disjoint union of two equally sized R'-orbits, and

$$|C_A| \geq 2(v_A + 1), \qquad |C_B| \geq v_B + 1.$$

If $v_A = v$, their total would be at least $2v + 3 > v + 3$, contradicting equality. Hence $v_B = v$, and equality forces $v_A = 0$, $|C_A| = 2$ and $|C_B| = v + 1$. The group R' fixes both points of C_A. Every further right mark fixes a and commutes with L, so also fixes its partner. Thus $A = C_A$ and R fixes A pointwise. On the other side, $b^{R'}$ already has at least $v + 1 = |C_B|$ points, so $C_B = b^{R'}$. The group L therefore sends b into its R-orbit. Commutation now gives $B = b^R$, with R transitive on B.

The crossing joins a to B. The other point z of the two-point side A is fixed by ρ_j: its sole possible partner in A is already used by the crossing, and a second crossing to B is forbidden by local uniqueness. Therefore $\langle \rho_j, R \rangle = Q_{>m}$ fixes z and is transitive on the complement. This proves both the full and truncated versions without assuming that the right action is symmetric. □

Retain now every actual odd-degree head and global paired-tail hypothesis, and choose the colour j by the complete simultaneous-witness theorem. Write $H = G_{\geq 5}$, $K = G_{>5}$ and $\sigma = \rho_4$. The first three marks are supported on D, and the acquired actual interface is

$$\rho_3 = (6\ b)\delta, \qquad H|_{D \cup \{b\}} = 1, \qquad \sigma|_D = 1.$$

Every K-intransitive witness orbit for j has a genuine local unique crossing and is fixed pointwise by δ.

Theorem 3. *Suppose two distinct witness orbits O_1, O_2 are K-intransitive, with last active labels $x < y$. If $x \geq j + 2$, then*

$$x < r - 1, \qquad |\operatorname{Fix}(G_{>x})| \geq 2x + 1.$$

Proof. The acquired two-witness theorem already gives $x < r - 1$ and at least $2x$ fixed points. Suppose the desired stronger bound fails, and put $F = \operatorname{Fix}(G_{>x})$. Then $|F| = 2x$. The complete count exhibited the following disjoint subsets of F: the seven points $D \cup \{b\}$, the whole orbit O_1 of size at least $x - 3$, and the two explicit truncated endpoint orbits C_A, C_B inside O_2, of total size at least $x - 4$. Consequently every bound is an equality, and, with $C = C_A \cup C_B$,

$$F = D \cup \{b\} \ \sqcup\ O_1 \ \sqcup\ C, \qquad |O_1| = x - 3, \qquad |C| = x - 4. \tag{1}$$

If $j \geq 7$, the complete minimum-orbit rigidity applies on O_1, since both side lengths are at least two; the truncated equality lemma applies on O_2. If $j = 5$ or 6, the short-left-side lemma applies in its full version on O_1 and its truncated version on O_2. Thus in every case the K-orbits on O_i are one fixed point z_i and a transitive complement T_i, with $z_2 \in C$. The mark $\rho_y \in K$ is nontrivial on T_2 but fixes T_1 pointwise.

The acquired actual anchor

$$z_0 = b^\sigma$$

is outside $D \cup \{b\}$ and fixed by K. It belongs to F, so (1) forces $z_0 = z_1$ or $z_0 = z_2$. The permutation σ centralizes K and hence permutes its actual orbits; it also preserves F, since it centralizes $G_{>x}$. We derive a contradiction in either case.

If $z_0 = z_i$ and σ preserves T_i, the proper set $D \cup \{b\} \cup O_i$ is invariant under every generator of G. The head preserves D; ρ_3 exchanges $6, b$ and fixes O_i through δ; σ exchanges b, z_i, fixes D and preserves T_i; and H preserves O_i and fixes D, b. The other witness orbit is outside this set, so this contradicts transitivity. Thus the complement associated with z_0 must be moved to a distinct K-orbit by σ.

If $z_0 = z_1$, every point of T_1 lies in F, so its distinct image orbit lies in F as well. It cannot meet the K-fixed head points or z_1, z_2. By (1) it must therefore meet T_2, and so equal T_2. This is impossible: the commuting bijection σ would conjugate $\rho_y|_{T_1} = 1$ to the nonidentity restriction on T_2.

If $z_0 = z_2$, choose a point in $C \setminus \{z_2\}$; this set is nonempty since $|C| = x - 4 \geq 3$. Its image lies in F and in the distinct orbit T_2^σ. It cannot be K-fixed and cannot lie in O_2, whose only nontrivial K-orbit is T_2. Equation (1) therefore places it in T_1. Hence $T_2^\sigma = T_1$, giving exactly the same contradiction with ρ_y. Both cases are impossible. Thus $|F| \geq 2x + 1$. □

Attribution and boundary. The complete simultaneous-witness count, minimum local orbit rigidity and actual head anchor are acquired. The new steps audit equality in the truncated count, cover the short left sides, and use the actual K-orbits and ρ_4-action on the entire fixed set. Every invariant-set assertion checks all generator families. The cases $x = j$ or $x = j + 1$, and a second witness with transitive K-action remain, as do full source routing and global attachments. No whole-original, question, numerical or formal credit is claimed.